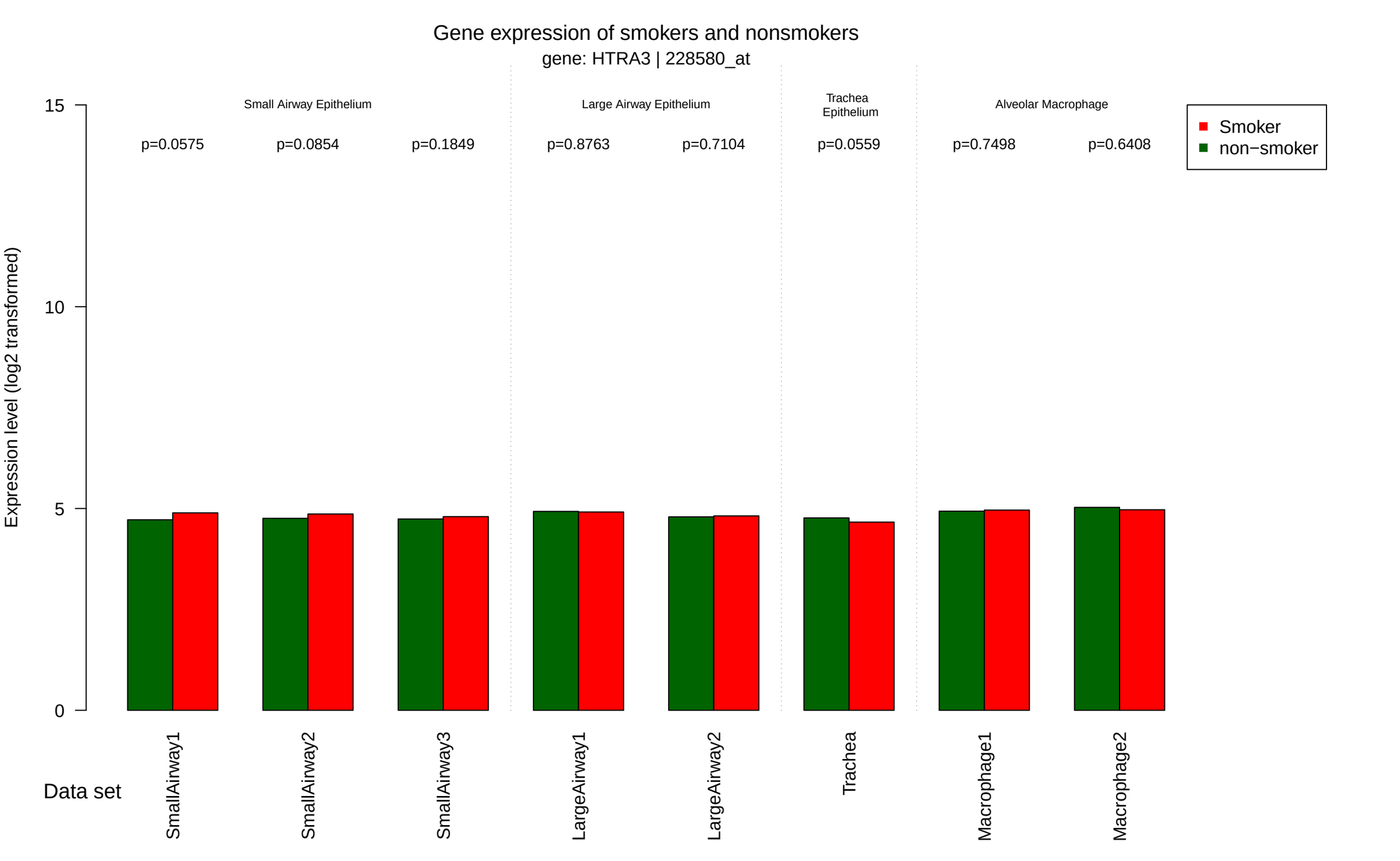
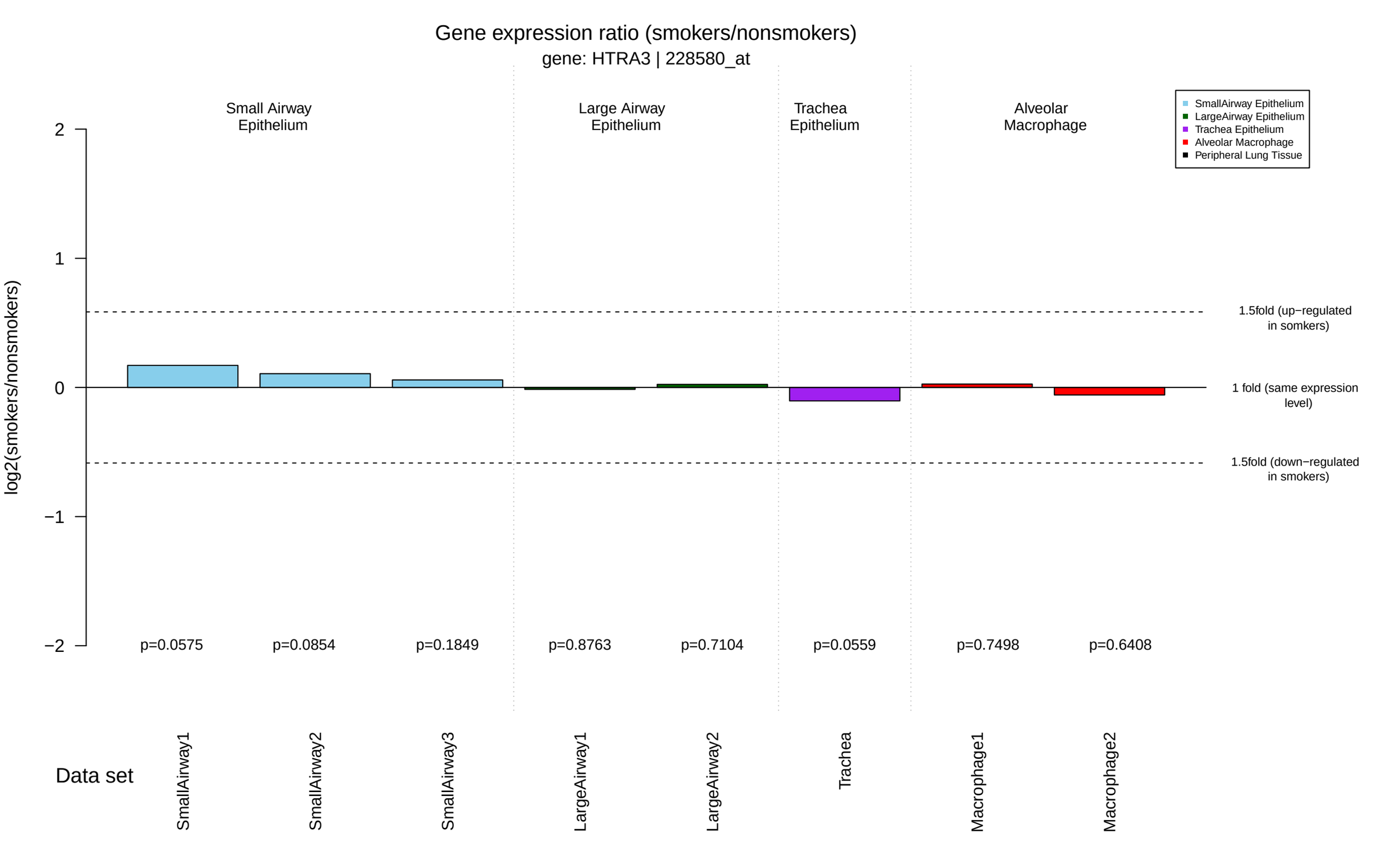
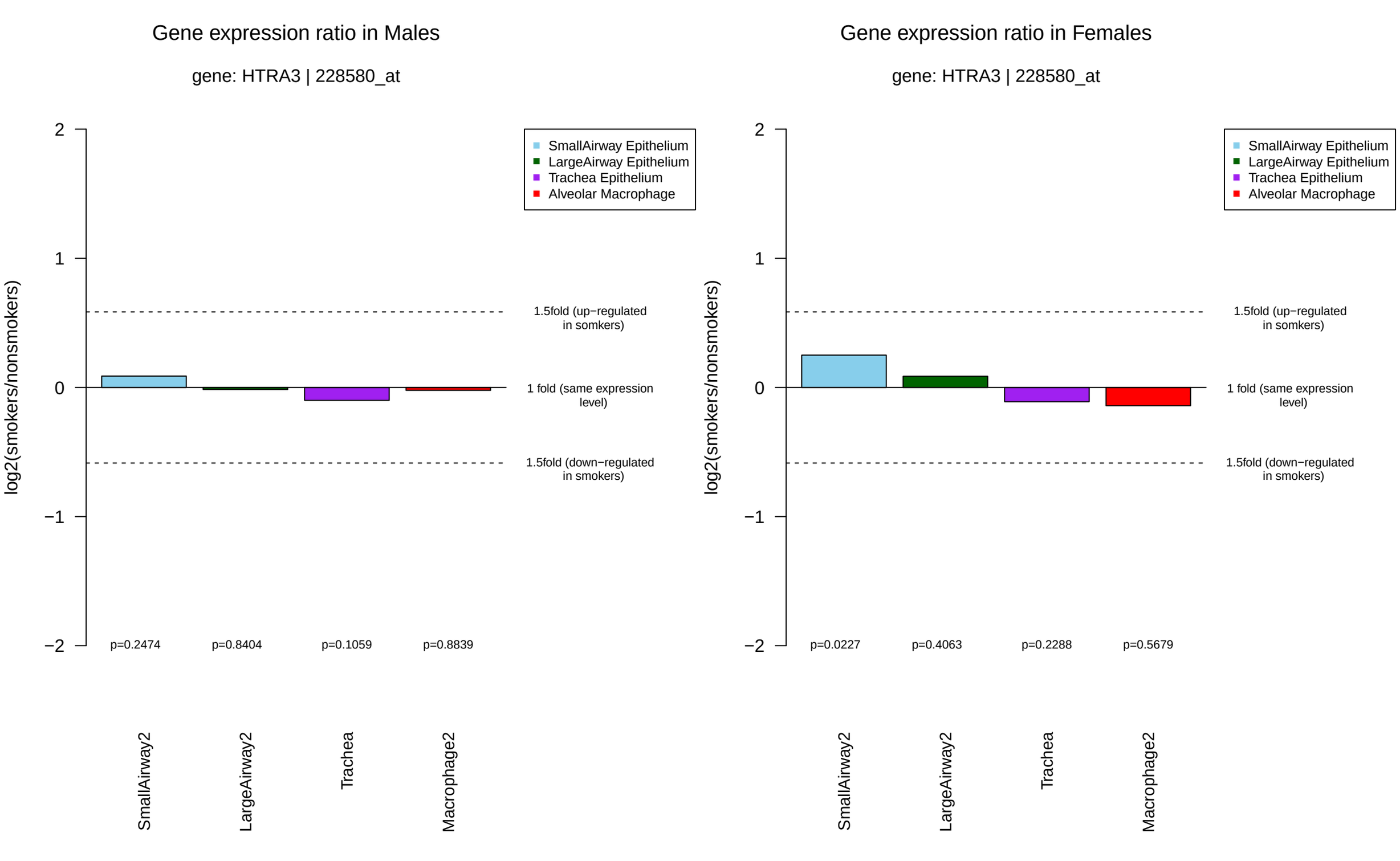
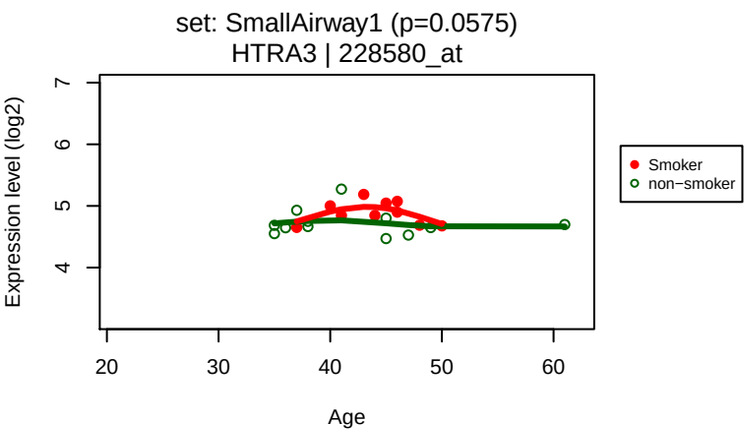
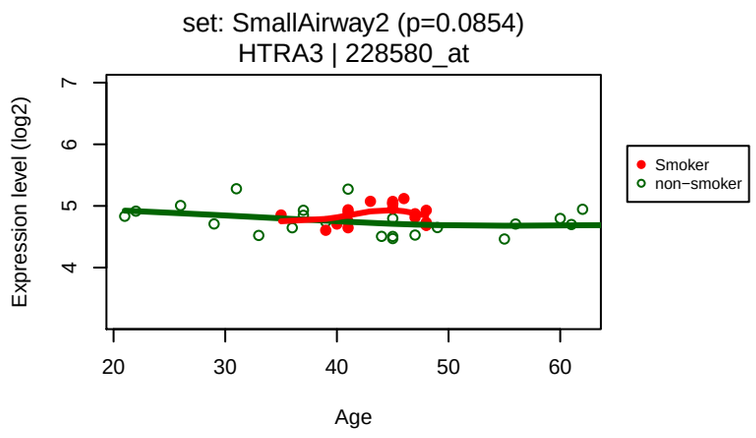
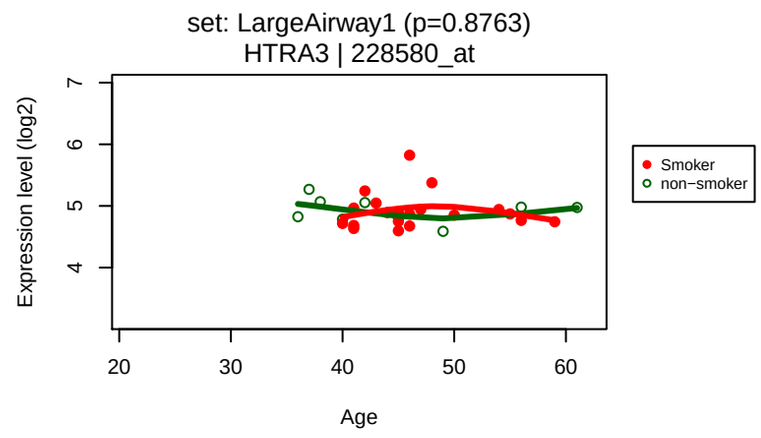

Gene expression of smokers and nonsmokers
gene: HTRA3 | 228580_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0575
p=0.0854
p=0.1849
p=0.8763
p=0.7104
p=0.0559
p=0.7498
p=0.6408
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

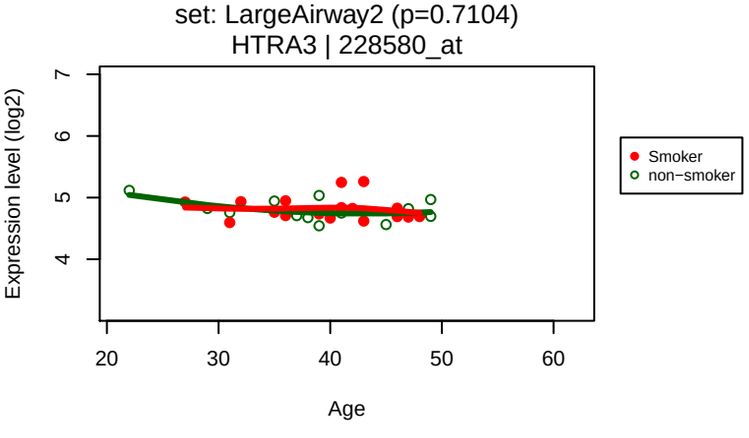

Gene expression ratio (smokers/nonsmokers)
gene: HTRA3 | 228580_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0575
p=0.0854
p=0.1849
p=0.8763
p=0.7104
p=0.0559
p=0.7498
p=0.6408
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

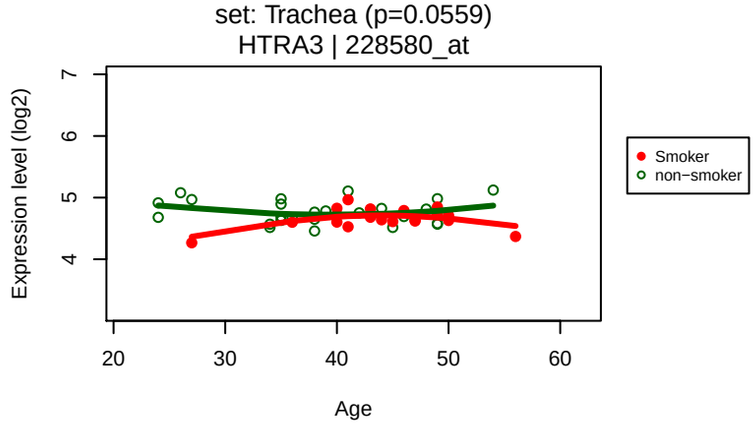

Gene expression ratio in Males
gene: HTRA3 | 228580_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2474
p=0.8404
p=0.1059
p=0.8839
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HTRA3 | 228580_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0227
p=0.4063
p=0.2288
p=0.5679
SmallAirway2
LargeAirway2
Trachea
Macrophage2

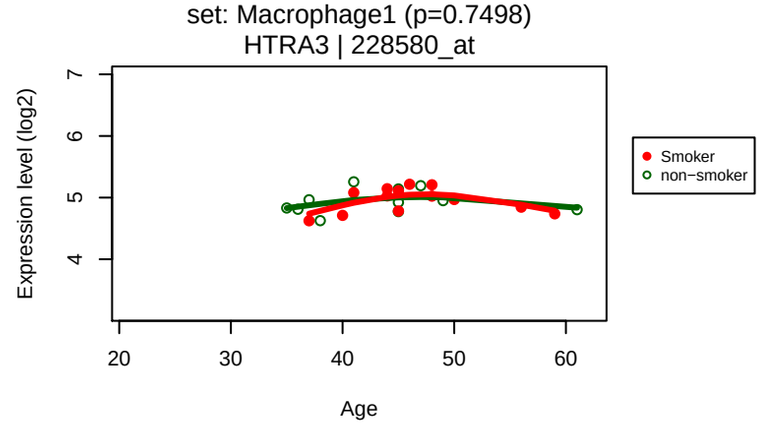

set: SmallAirway1 (p=0.0575)
HTRA3 | 228580_at
Expression level (log2)
Age
Smoker
non−smoker

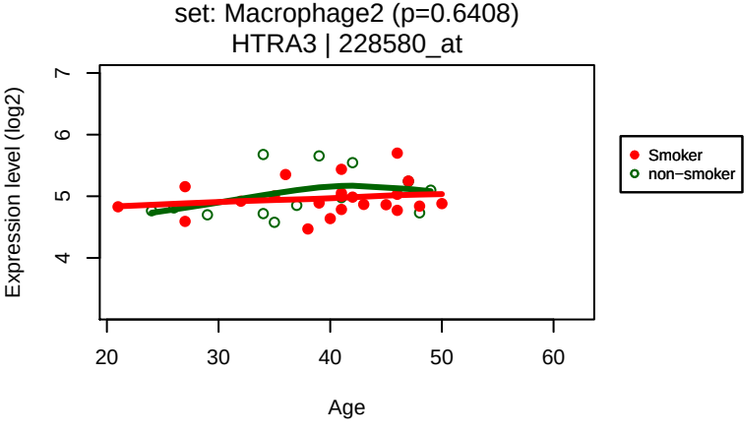

set: SmallAirway2 (p=0.0854)
HTRA3 | 228580_at
Expression level (log2)
Age
Smoker
non−smoker

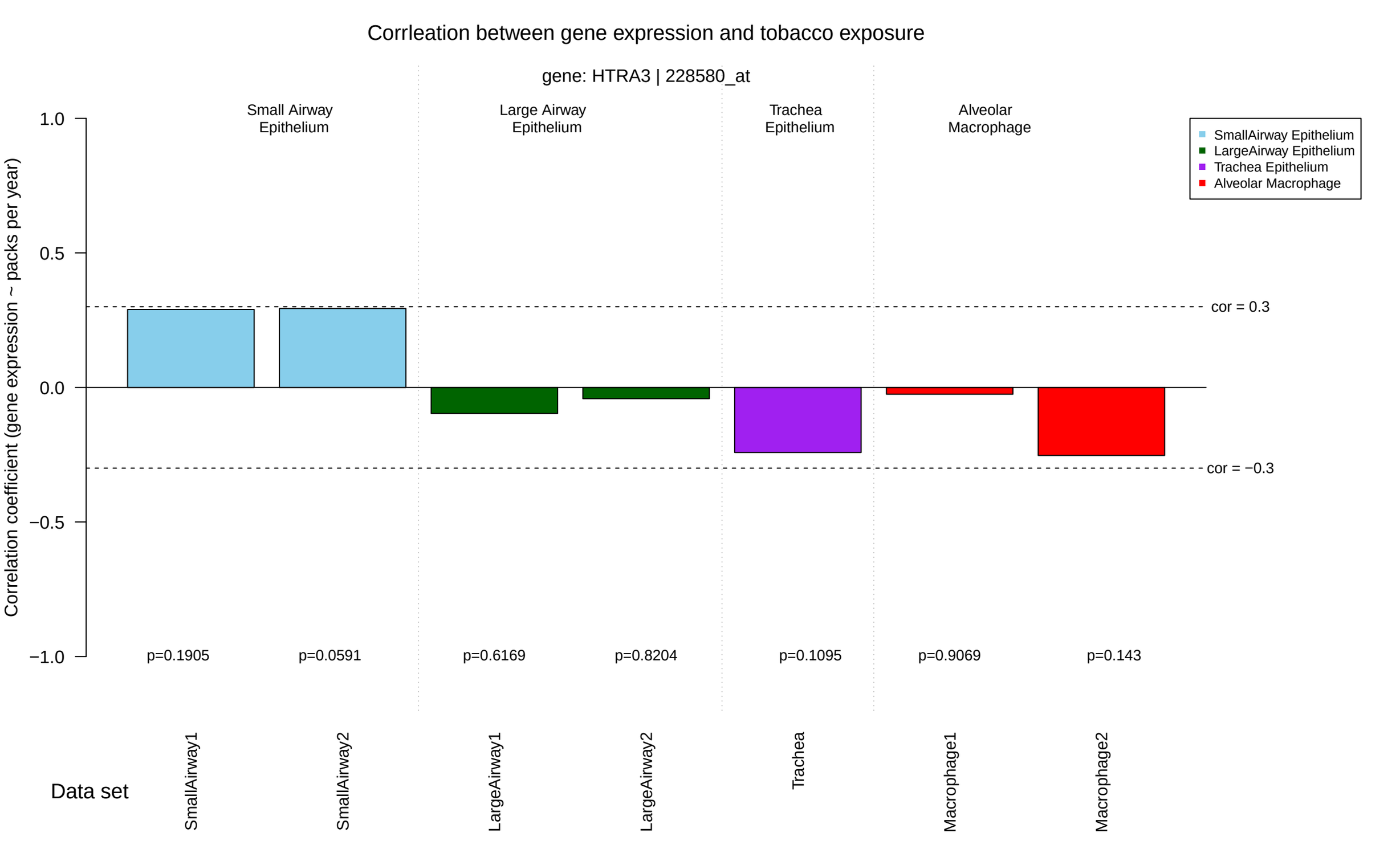

set: LargeAirway1 (p=0.8763)
HTRA3 | 228580_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7104)
HTRA3 | 228580_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0559)
HTRA3 | 228580_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7498)
HTRA3 | 228580_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6408)
HTRA3 | 228580_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HTRA3 | 228580_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1905
p=0.0591
p=0.6169
p=0.8204
p=0.1095
p=0.9069
p=0.143
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2